Task 09: Evaluate the Memory Hierarchy and I/O Systems

Operating Systems (CSA04) — Assignment 1 (CO1)

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Marks: 4 Marks (Bloom's: K5 - Evaluate)
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> ***Task 9 Description:*** *Evaluate the memory hierarchy (registers to cloud storage). Define data placement for MRI scans, EHR records, and monitor readings. Compare Programmed I/O, Interrupt-driven I/O, and DMA. Quantify CPU overhead when transferring a 500 MB MRI scan via interrupts vs. DMA.*

1. MEMORY HIERARCHY SPECIFICATION & PLACEMENT PLAN

Storage Tier	Access Latency	Capacity	Hospital Data Placement & Rationale
CPU Registers	~0.5 – 1.0 ns	< 2 KB	Active ECG filtering variables, sensor register copies, alarm threshold values.
L1 / L2 / L3 Caches	~1 – 15 ns	32 KB – 64 MB	3D FFT reconstruction coefficients and active database B-tree index nodes.
Main Memory (DDR5 RAM)	~60 – 80 ns	16 GB – 512 GB	Active patient telemetry circular buffers, DBMS buffer pool, uncompressed MRI arrays.
Solid State Drive (NVMe)	~10 – 25 µs	2 TB – 30 TB	Database write-ahead logs (WAL), completed DICOM images, operating system images.
Network / Cloud Storage	~10 – 50 ms	Petabytes	Archived patient histories, shared partner hospital records, offsite disaster backups.

2. COMPARISON OF I/O TECHNIQUES

- **Programmed I/O (PIO):** The CPU polls the device status register in a tight loop and moves data word-by-word through registers. *Rejected:* Wastes 100% of CPU cycles during data transfers.
- **Interrupt-Driven I/O:** The device sends a hardware interrupt whenever a word is ready. The CPU suspends work to run an ISR, moves the word to RAM, and resumes. *Rejected for high-throughput imaging:* High interrupt volume saturates the CPU.
- **Direct Memory Access (DMA) — Adopted:** The CPU configures the DMA controller with source, destination, and byte count, then initiates the transfer. The DMA controller transfers data directly between the scanner and RAM over PCIe buses, generating a **single interrupt** once the transfer completes.

3. QUANTITATIVE ANALYSIS: 500 MB MRI TRANSFER (INTERRUPTS VS. DMA)

Transferring a 500 MB (500×10^6 bytes) MRI image on a 64-bit architecture (8 bytes/word):

- **Interrupt-Driven Transfer:**
  $\text{Total Interrupts} = (500 \times 10^6) / 8 = 62{,}500{,}000$ interrupts.
  Assuming an ISR context switch time of 1.5 µs (1.5×10^{-6} s):
  $\text{Total CPU Time} = 62{,}500{,}000 \times 1.5 \times 10^{-6} = 93.75$ seconds.
  The CPU is fully saturated for over 1.5 minutes, dropping real-time sensor updates and freezing other applications.
- **DMA Transfer:**
  The CPU sets up the transfer and receives **1 single interrupt** upon completion.
  $\text{Total CPU Time} = \sim 2$ microseconds.
  The 500 MB transfer completes across PCIe Gen 4 buses at line rates (> 10 GB/s) in under 0.05 seconds while the CPU continues image reconstruction tasks concurrently.